

The line was divided into seven sectors, each about 100 km, headed by chief engineers with enough authority to allow them to cut through the bureaucratic red tape. Sreedharan believed that five years was enough time to complete a 100–120-km-line. Additionally, there was a small team of experienced officers who provided their expertise to the different sectors with 400 freshly recruited engineers on the ground. These young survey engineers were given motorcycles, modified to carry survey equipment, to travel across Goa quickly to work on alignments and collect data. The organization remained small with about 2,400 people in its employ at its peak.

The construction

Laying the 738.9-km-broad-gauge line across three states, while crossing the Sahyadri Hills was a formidable task with rocky hills, treacherous soil conditions, innumerable rivers, big and small, and an intense annual monsoon added to the mix. "The saga of the construction of Konkan Railway is one of boring, drilling, cutting, and dynamiting your way through (the) arms and tentacles of the Western Ghats that came in the path of the rail line," wrote Joseph George, the general manager of the KRCL in an article.

Today, the line with 342 curves, crosses 67 stations across 1,880 bridges, the longest of which is across the Sharavati River in Honnavar, Karnataka at 2,065.8 m. Maharashtra's Panval Nadi is the tallest viaduct, nearly 423-m high, built using prestressed concrete girders to cross a ravine, as was the norm for this line. The viaducts used incremental launching by building the bridge from one end and pushing it forward, in a first for India. Welded steel triangular girders were also used for the bridges built over water. The tracks laid could accommodate high-speed trains that run at 160 km/h.

Building tunnels

The greatest challenge however lay in the tunnels, 91 in total, with the Karbude tunnel near Ratnagiri in Maharashtra as the longest at 6.5 km. The problem arose between Sawantwadi and Canacona in Goa where soil conditions threw engineers for a loop as did constant roof collapses